P3 | Axial-piston pump

Compact, water-lubricated axial-piston technology for a defined operating duty.

P3 supplies pressurised water to hydraulic actuators or a process. Select delivered flow at the working pressure, then match displacement, speed, drive power and inlet supply.

Published parameter	P3 range rating
Maximum displacement (cm³/rev)	3.0
Minimum displacement (cm³/rev)	3.0
Maximum speed, unboosted (rpm)	1800
Maximum speed, boosted (rpm)	2000
Maximum input power (kW)	1.75
Maximum water flow (L/min)	6.0
Maximum continuous pressure (bar)	160
Mass (kg)	1.9
Maximum temperature (°C)	50
Minimum temperature (°C)	2

Nominal ordering displacements: 3 cm³/rev. Nominal options and range-table limits are different source entries; use the selected configuration.

Design & environmental advantages

Multiple pistons smooth the delivery of power. Water lubricates the working surfaces, removing hydraulic oil from this circuit. Standard stainless construction and selectable internal materials support fluid-specific selection.

Efficiency & selection

Match component size to useful work. Read the supplied efficiency curve at pressure and speed. Avoid sustained relief flow and excessive throttling; these convert useful hydraulic power into heat.

Installation and procurement

Specify flow, pressure, speed, fluid chemistry, inlet supply, rotation, seal type, shaft coupling and mounting. No induced radial or axial shaft load. Nominal 10 µm / 25 µm absolute filtration, beta10 ≥ 75, is listed in the pump catalogue. Variable versions are listed for P60 and P180 only.

Maxima are separate catalogue limits. Obtain the exact configuration drawing, duty-point selection, seal/service schedule, material documentation and applicable approval evidence with the quotation.



Family photograph; configuration varies.

Connect the selected design

Two-port pumps need flooded, positive inlet pressure and cannot lift water. Three-port boosted units separate the feed and case drain. Specify that arrangement in the order.

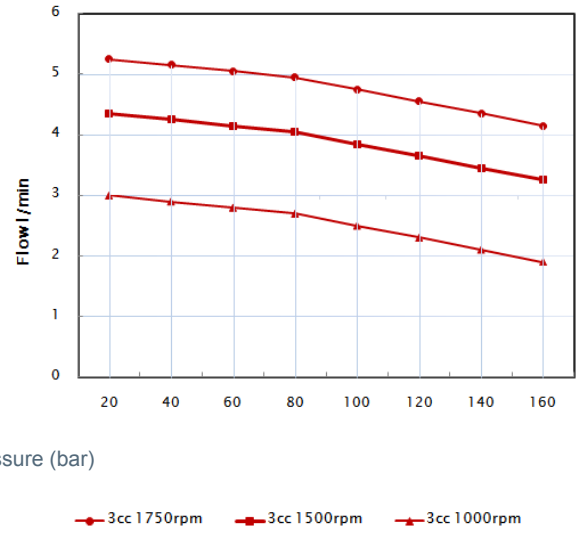
Reference conversions: 160 bar = 2,321 psi; 1 L/min = 0.264172 US gpm; 1 kg = 2.20462 lb.

The Water Hydraulics Co. Janus pump catalogue V.01.25, pp. 2-4. Website product context checked 17 Oct 2026.

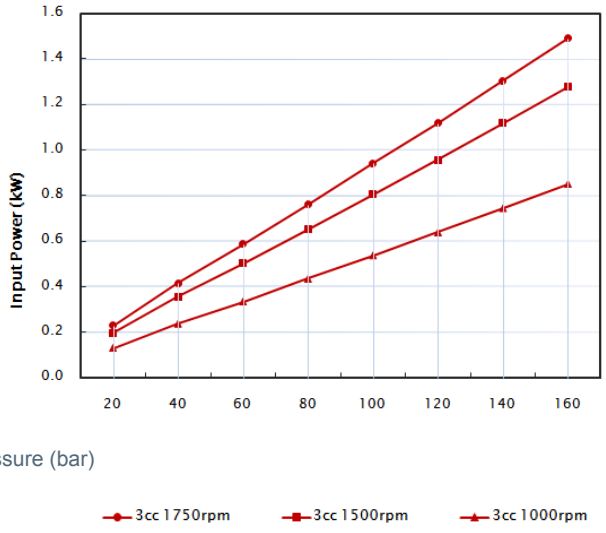
P3 | Performance curves

Read displacement, speed and pressure together. Published curve data are preserved.

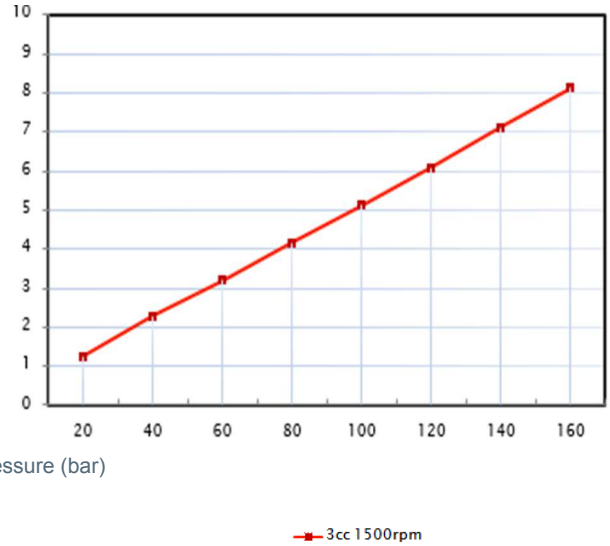
Output flow (L/min)



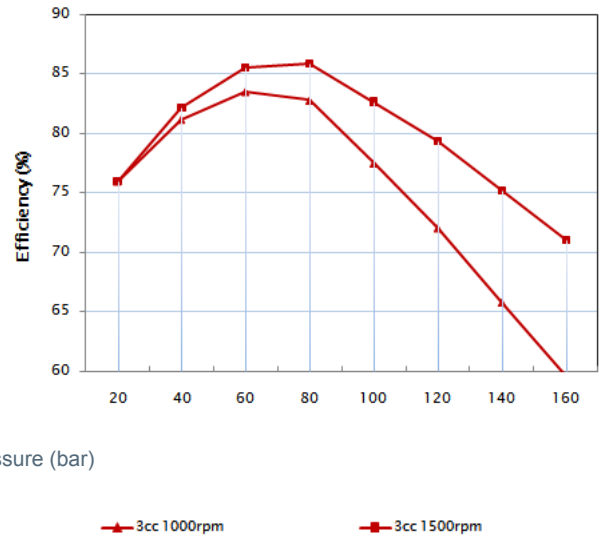
Input power (kW)



Input torque (N·m)



Overall efficiency (%)



Apply the curves to the duty

Published curves are operating references, not simultaneous maximum ratings. Read efficiency at the proposed duty instead of quoting a peak value for the complete system.

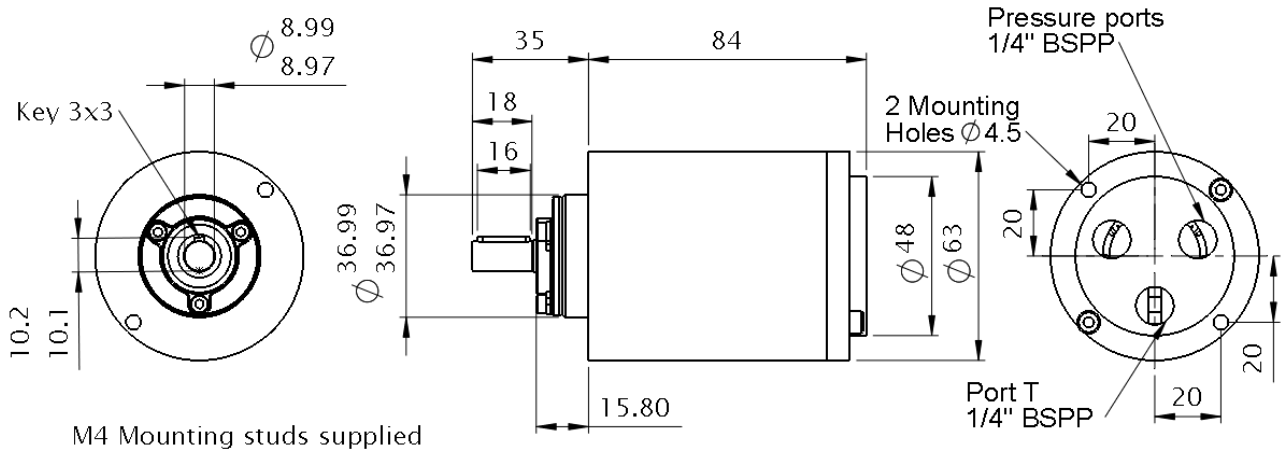
1 L/min = 0.264172 US gpm; 1 bar = 14.5038 psi; 1 kW = 1.34102 hp; 1 N·m = 0.737562 lbf·ft.

The Water Hydraulics Co. Janus pump catalogue V.01.25, p. 8.

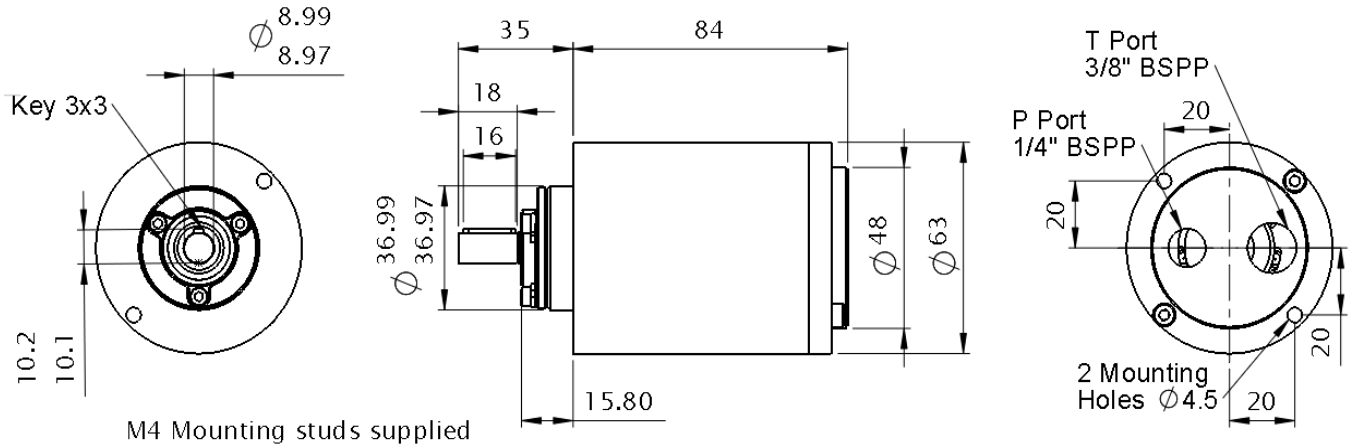
P3 | Dimensions & connections

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

PB160-3.0W



PC160-3.0W & PA160-3.0W



Port threads and flange classes are connection designations, not converted diameters or operating-pressure ratings. Select the exact configuration shown on the approved order drawing.

The Water Hydraulics Co. Janus pump catalogue V.01.25, p. 7.